L01: Algorithms & Analysis

Asymptotic bounds & sums

$O(g) : f \le Cg; \Omega(g) : f \ge cg;$
$\Theta(g) : cg \le f \le Cg\ (0 < c \le C, n \ge n_0).$
$\Theta(g) = O(g) \cap \Omega(g)$; bounds $\neq$ best/worst input cases.

$$1 \prec \log n \prec n \prec n \log n \prec n^2 \prec 2^n \prec n!,$$
$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} = \Theta(n^2),$$
$$\sum_{i=0}^{h} r^i = \frac{r^{h+1}-1}{r-1}\ (r \neq 1),$$
$$H_n = \sum_{i=1}^{n} \frac{1}{i} = \Theta(\log n),$$
$$\sum_{i=1}^{n} \log i = \log(n!) = \Theta(n \log n).$$

Loop templates: $\sum_{t=0}^{\lfloor \log_b n \rfloor} b^t = \Theta(n)\ (b > 1)$; $\sum_{t=0}^{\lfloor \log_b n \rfloor} \lfloor n/b^t \rfloor = \Theta(n)$; $\sum_{i=1}^{h} (\lfloor \log_b i \rfloor + 1) = \Theta(n \log n)$. Θ **proof:** upper by max term; lower from last $\ge n/2$ terms (each $\Omega(\log n)$ for log sums).

Correctness & searching

Loop invariant: initialization $\rightarrow$ maintenance $\rightarrow$ termination; also prove progress/termination. **Linear:** best $\Theta(1)$; worst $\Theta(n)$; successful uniform average comparisons $(n+1)/2$; space $\Theta(1)$. **Binary:** sorted A; best $\Theta(1)$, worst/standard average $\Theta(\log n)$; iterative aux. space $\Theta(1)$, recursive (index-only) stack $\Theta(\log n)$.

$$m = l + \left\lfloor \frac{r-l}{2} \right\rfloor,$$
$$\begin{cases} l > r & \Rightarrow \text{return } -1, \\ A[m] = x & \Rightarrow \text{return } m, \\ A[m] < x & \Rightarrow l \leftarrow m+1, \\ A[m] > x & \Rightarrow r \leftarrow m-1. \end{cases}$$

Invariant: if x exists, it remains in $A[l \ldots r]$; $l = r$ is one candidate, $l > r$ is empty (do not index).

Sorting

Algorithm	Best	Worst	Aux.	Stable
Bubble V4	n	n^2	1	Y
Insertion	n	n^2	1	Y
Selection	n^2	n^2	1	N
Merge	$n \log n$	$n \log n$	n	Y*

*Entries are $\Theta(\cdot)$; *merge ties: take left first.* **Invariants:** Bubble: suffix contains i largest (sorted); Insertion: processed prefix sorted; Selection: prefix contains i smallest (sorted); Merge: output prefix sorted after each extraction.

$$I = \sum_{i<j} [A_i > A_j], \qquad T_{\text{ins}} = \Theta(n + I),$$
$$C_{\text{sel}} = \frac{n(n-1)}{2},$$
$$T_{\text{merge}} = 2T(n/2) + \Theta(n) = \Theta(n \log n).$$

Special: ordered segments of length k: bubble $\Theta(nk)$ worst case, at most $k-1$ whole-array passes (known condition); $[2, 3, \ldots, n, 1]$: bubble $\Theta(n^2)$, insertion $\Theta(n)$. Two distinct keys can still give $I = \Theta(n^2)$. Selection's nonadjacent swap may break stability; Bubble V4 stops on a no-swap pass. Standard merge peak aux. space $\Theta(n)$ (reused buffer); L01 slides list $O(n \log n)$ for a different allocation/accounting convention.

Recursion tree & Master Method

For $T(n) = aT(n/b) + f(n)\ (a \ge 1, b > 1)$, let $p = \log_b a$, $h = \log_b n$ (assume $n = b^h$, $T(1) = \Theta(1)$):

$$W_i = a^i f(n/b^i), \quad L = a^h = n^p,$$
$$T(n) = \sum_{i=0}^{h-1} W_i + \Theta(n^p).$$

For $f(n) = \Theta(n^c)$: $W_i = \Theta\left(n^c (a/b^c)^i\right)$; $c < p$: leaves; $c = p$: equal levels; $c > p$: root. **Basic Master** (eventually nonnegative f, $\epsilon > 0$):

$$f = O(n^{p-\epsilon}) \Rightarrow T = \Theta(n^p),$$
$$f = \Theta(n^p) \Rightarrow T = \Theta(n^p \log n),$$
$$f = \Omega(n^{p+\epsilon}) \Rightarrow T = \Theta(f(n)).$$

Case 3 also requires $a f(n/b) \le \rho f(n)$, $\rho < 1$.

Extended Master: $f = \Theta(n^c \log^k n)$ (eventually positive):

$$T(n) = \begin{cases} \Theta(n^p), & c < p, \\ \Theta(n^c \log^k n), & c > p, \\ \Theta(n^p \log^{k+1} n), & c = p,\ k > -1, \\ \Theta(n^p \log\log n), & c = p,\ k = -1, \\ \Theta(n^p), & c = p,\ k < -1. \end{cases}$$

At $c = p$, $h = \log_b n$:

$$T = \Theta\left(n^p \sum_{j=1}^{h} j^k\right).$$

$\sum j^{-1} = \Theta(\log h)$, $\sum j^{-2} = \Theta(1)$.

L01 exercise traps

Signed recurrence: standard Master does not apply to $f(n) = n \sin n$. For $n = 2^h$, $T(1) = c > 0$:

$$T(n) = n^2 \left(c + \sum_{j=1}^{h} \frac{\sin(2^j)}{2^j}\right) = \Theta(n^2).$$

Indeed $S_2 = \sin 2/2 + \sin 4/4 \approx 0.26545 > 1/4 = \sum_{j \ge 3} 2^{-j}$, hence $0 \le S_h \le 1$. Negative terms can cancel: $4T(n/2) - n$, $T(1) = 1$ has solution $T(n) = n$.